

MODEL QUESTION PAPER

TED (10) 5040

Revision 2010

FIFTH SEMESTER DIPLOMA EXAMINATION IN ENGINEERING/TECHNOLOGY

SUB: ADVANCED MICROPROCESSORS

Part A

I. Answer all questions in one or two sentences. Each carries two marks.

1. Give the maximum memory size that can be addressed by 8086.
2. Mention what happens when the interrupt bit of flag register in 8086 is kept in set condition.
3. Define the term hardware interrupt.
4. List any two Intel microprocessors released after the year 2000.
5. Expand the terms CISC and RISC.

Part B

II. Answer any five questions. Each carries 6 marks.

- a. Write an ALP to find the sum of ten 8 bit numbers.
- b. Write any six data transfer type instructions in 8086.
- c. List the three different sources of interrupts in 8086.
- d. Mention the features of programmable interrupt controller 8259.
- e. Differentiate real address mode and protected virtual address mode in 80286.
- f. List any six differences between 80286 and 80386.
- g. Write the concept of super scalar architecture in Pentium processor.

Part C

Answer one full question from each unit. Each carries fifteen marks

Unit- I

- III. Draw and explain the architecture of 8086. (15)

OR

- IV. Explain various addressing modes of 8086 with one example each. (15)

Unit- II

- V. With a neat diagram, explain how Intel 8279 is interfaced with 8086. (15)

OR

- VI. a) Describe the interrupt vector table in 8086. (10)
b) Briefly discuss the various steps involved in DMA operation in 8086 (5)

Unit- III

VII. Illustrate the architecture of 80286. (15)

OR

VIII. Explain the different modes of operation in 80386. (15)

Unit IV

IX. Draw and explain the Pentium architecture.

OR

X. a) Compare Pentium II with Pentium III processor. (10)

b) Give a brief description of data and instruction cache in Pentium. (5)
